

SEMESTER END EXAMINATIONS – FEB / MARCH 2025

Program	: B.E. – Information Science and Engineering	Semester	: V
Course Name	: Software Engineering	Max. Marks	: 100
Course Code	: IS53(000) (Prior to 2021 Batch)	Duration	: 3 Hrs

Instructions to the Candidates:

- Answer one full question from each unit.

UNIT - I

- Illustrate the concepts of software engineering for an information system used to manage and provide access to a database. CO1 (10)
 - Discuss with diagram the RUP process model from a dynamic perspective, elaborating the phases. CO1 (10)
- Compare and contrast agile development and plan-driven software development. CO1 (10)
 - What is a software process model? Discuss the fundamental activities carried out in the incremental development. CO1 (10)

UNIT - II

- With the diagram explain the types of non-functional requirements. CO2 (08)
 - Discuss briefly the structure of a requirements document. Also, discuss ways of writing a System Requirements Specification. CO2 (06)
 - Discuss the different checks and techniques used in requirements validation. CO2 (06)
- Identify and briefly describe four types of requirement that may be defined for a computer based system. CO2 (06)
 - What is requirements discovery? Explain the requirement discovery process by considering an example of your choice. CO2 (08)
 - With the diagram explain the principal stages to a change management process. CO2 (06)

UNIT - III

- What do you mean by a software architectural pattern? Describe with a diagram the organization of the Model-View-Controller pattern. CO3 (10)
 - Explain any five design principles used while designing class-based components. CO3 (10)
- Discuss the software architecture of an ATM system. Illustrate the functions of the input, process and output components. CO3 (10)
 - Discuss the activities involved in developing an object-oriented system design from concept to detail. Are these activities to be done sequentially? CO3 (10)

UNIT- IV

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| 7. | a) | Define the following terms: Fault, Error, Fault, Incident, Test, Test-cases. | CO4 | (06) |
| | b) | Develop a formula for the number of robust worst-case test cases for a function of n variables. | CO4 | (08) |
| | c) | Discuss the equivalence class test cases for the NextDate function. | CO4 | (06) |
| 8. | a) | Make a Venn diagram that reflects a part of the following statement: "... we have left undone that which we ought to have done, and we have done that which we ought not to have done ...". | CO4 | (06) |
| | b) | List any four Normal Boundary Value Test Cases for triangle problem. | CO4 | (06) |
| | c) | Illustrate with an example an Improved Equivalence Class testing. | CO4 | (08) |

UNIT - V

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| 9. | a) | What is risk identification? Discuss different types of risks that may be included in a risk checklist with examples. | CO5 | (10) |
| | b) | Discuss in detail the four critical factors used in people management. | CO5 | (10) |
| 10. | a) | Highlight any six software quality attributes. Explain any two. | CO5 | (10) |
| | b) | Discuss the factors affecting software pricing. | CO5 | (10) |
